EC2 Deployment Essentials

1. Instance Types

Amazon EC2 offers various types of virtual servers (instances), each built for different workloads.

- **General Purpose (e.g., t3, t4g)** – Balanced for everyday apps and small servers.
- **Compute Optimized (e.g., c5)** – Great for CPU-heavy tasks like batch processing.
- **Memory Optimized (e.g., r6g)** – Ideal for large databases and in-memory apps.
- **Storage Optimized (e.g., i4i)** – Best for apps needing high disk performance.
- **Accelerated Computing (e.g., p4, g4dn)** – Used for AI/ML, video rendering, and simulations.

EC2 Instance Types:

Type	Use Case	Examples
General Purpose	Balanced CPU, memory, and networking	`t3`, `t4g`, `m6g`
Compute Optimized	High-performance CPU workloads (e.g., gaming, ML)	`c6g`, `c5`
Memory Optimized	RAM-intensive tasks (e.g., DBs, analytics)	`r6g`, `x2idn`
Storage Optimized	High-speed, high-throughput storage apps	`i4i`, `d3en`
Accelerated Compute	GPU/AI, 3D graphics, machine learning	`p4`, `g4dn`, `f1`

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2. AMI (Amazon Machine Image)

AMI acts like a template for launching an instance with pre-installed OS and software.

- **Free-tier AMIs** (like Amazon Linux, Ubuntu) are free and basic.
- **Licensed AMIs** (e.g., Windows Server, Red Hat) include pre-configured apps and incur cost.
- You choose AMI during instance launch and can also create your custom AMIs.

AMI Types:

AMI Type	Details
Free-Tier	Basic OS images (Amazon Linux, Ubuntu) — No license cost
Licensed AMIs	Comes with pre-installed software (Windows Server, Red Hat)
Custom AMIs	User-created from an existing instance — speeds up deployments

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3. Security Groups

Security Groups control the flow of traffic to and from your EC2 instance (like a firewall).

- **Inbound Rules** allow specific ports (e.g., 22 for SSH, 80 for HTTP).
- **Outbound Rules** manage what traffic your instance can access.
- They are **stateful** – if inbound is allowed, response traffic is automatically allowed.

Security Groups (Firewall)

Component	Description
Inbound Rules	Control allowed traffic **into** the instance (e.g., SSH, HTTP, RDP)
Outbound Rules	Control traffic **leaving** the instance (e.g., internet access)
Stateful	Automatically allows return traffic if the incoming rule permits
Usage	Attach to instances, modify anytime, can be reused across instances

4. EBS (Elastic Block Store) Types

EBS provides persistent storage to your EC2 instances.

- **gp3 (General Purpose SSD)** – Default, fast, and low-cost.
- **io1/io2 (Provisioned IOPS SSD)** – High-performance for critical databases.
- **st1/sc1 (HDD Types)** – Low-cost for big, sequential data (logs, backups).
- **Magnetic** – Older generation, now rarely used.
  You attach volumes during or after launching the instance.

EBS Volume Types:

Volume Type	Use Case	Performance/Cost
gp3 (SSD)	General workloads, fast, free-tier eligible	Balanced, scalable IOPS
io1 / io2 (SSD)	High-performance DBs, critical apps	Expensive but very high IOPS
st1 (HDD)	Big data, streaming logs	Throughput optimized, cheaper
sc1 (HDD)	Rarely accessed data, backups	Lowest cost, slowest
Magnetic (standard)	Legacy use only	Deprecated, avoid for new apps

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Exploring AWS Console (Read-Only View)

5. EC2 Dashboard

The main EC2 screen gives access to all instance-related features.

- You can **launch instances**, **monitor status**, manage **storage**, **security groups**, and **Elastic IPs**.
- Launch flow: Choose AMI → Choose instance type → Configure → Add storage/tags → Review & launch.

EC2 Dashboard Components:

Section	Purpose
Instances	View, start/stop, terminate, and monitor instances
Launch Instance	Step-by-step wizard to deploy an EC2 instance
Volumes	Attach, detach, expand EBS volumes
Security Groups	Create/modify rules for inbound and outbound traffic
Elastic IPs	Reserve and associate static public IPs

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6. VPC (Virtual Private Cloud)

VPC allows you to set up a secure, isolated network inside AWS.

- **Default VPC** is created automatically — with public subnets and internet access.
- **Custom VPC** lets you design your own IP ranges, subnets, routing, and security controls.
- It's essential for enterprise-grade architecture and isolation.

VPC (Virtual Private Cloud):

VPC Type	Description
Default VPC	Automatically provided; ready-to-use with public subnet
Custom VPC	User-defined subnets, route tables, ACLs, NAT Gateways for control
Subnets	Divides VPC into public/private zones
Use Case	Isolate environments, restrict access, secure backend services

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7. Security Group Configurations

- Found inside VPC settings or EC2 console.
- You can define access per port, per IP (e.g., only allow SSH from office IP).
- For multi-tier apps, use multiple security groups (e.g., frontend, backend, database).

Security Group Configuration (Example):

Tier	Port	Access From	Protocol	Purpose
SSH (Admin)	22	Admin IP only	TCP	Secure terminal access
HTTP (Web)	80	Anywhere (0.0.0.0/0)	TCP	Public web server access
HTTPS (Secure)	443	Anywhere	TCP	Encrypted web server access
RDP (Windows)	3389	Admin IP	TCP	Remote Desktop access (Win)

Key Pairs (.pem vs .ppk)

8. Key Pair Purpose

- AWS uses key pairs for securely connecting to EC2 instances without passwords.
- The **private key** stays with you (.pem or .ppk file), and AWS stores the public key.

Key Pair Formats:

Format	Used With	Platform	Purpose
`.pem`	OpenSSH CLI	Linux/macOS	SSH access from terminal
`.ppk`	PuTTY, WinSCP	Windows	Connect using GUI-based SSH tools
Note	`.pem` is default from AWS. Use PuTTYgen to convert `.pem` to `.ppk` if needed		

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9. .pem vs .ppk Format

- **.pem** – Used in Linux/macOS terminal for SSH.
- **.ppk** – Required by PuTTY or WinSCP on Windows.
- AWS gives you a .pem by default; convert it to .ppk if using PuTTY.

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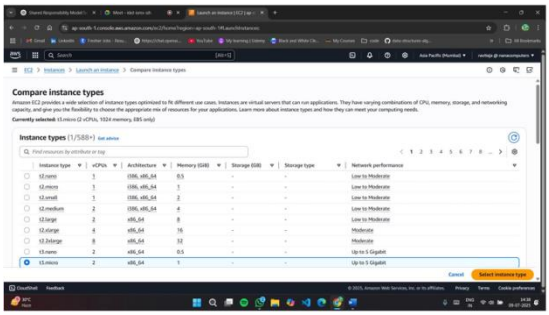
10. Convert .pem to .ppk (Using PuTTYgen)

- Open **PuTTYgen** (installed with PuTTY).
- Click **Load** > Select your `.pem` file (set file type to *All Files*).
- Click **Save Private Key** – This creates a `.ppk` file.
- Use `.ppk` to log into EC2 via PuTTY.

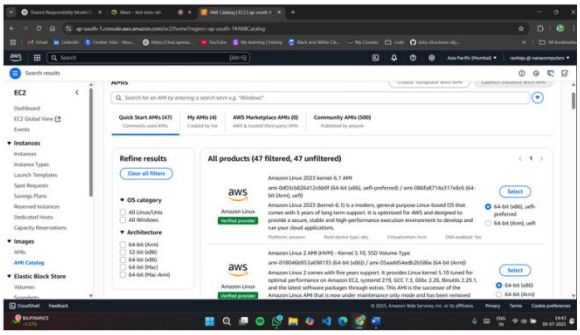
Converting `.pem` to `.ppk`:

Step	Description
Open PuTTYgen	Installed with PuTTY
Click **Load**	Select your `.pem` file (set file filter to all)
Click **Save Private Key**	Save as `.ppk` file
Use with PuTTY or WinSCP	Login to EC2 on Windows

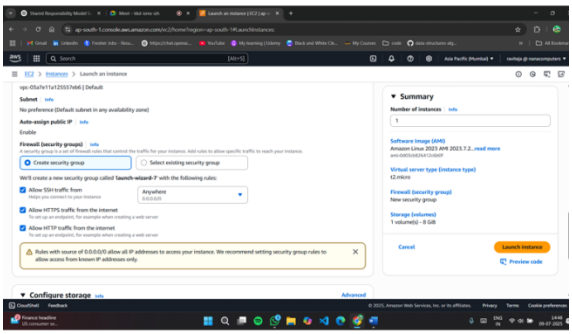
SCREENSNAPS :



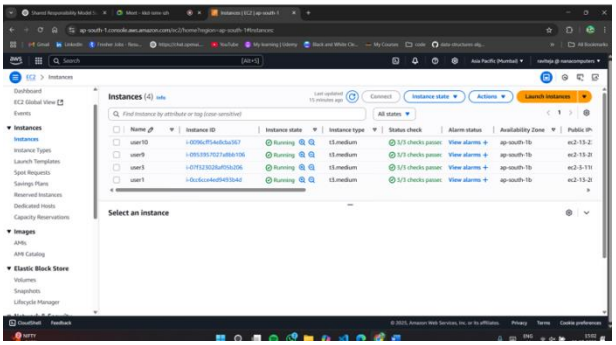
INSTANCE TYPES



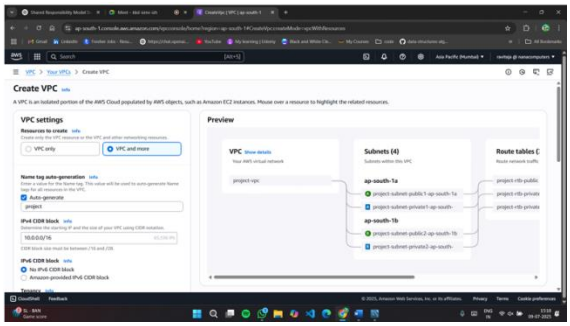
AMI'S



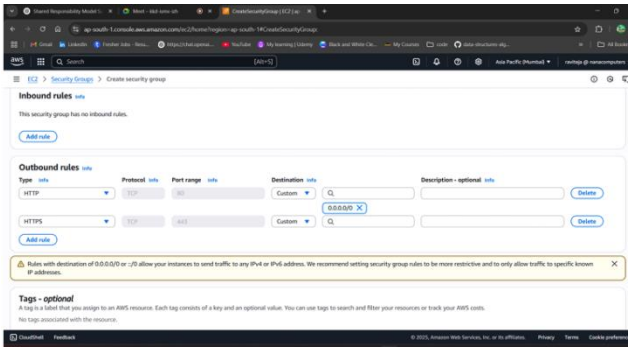
SECURITY GROUPS



AWS CONSOLE



VPC



INBOUND & OUTBOUND